Worksheet: Proof Techniques (Topic 2, Part 1)

Introduction

Welcome to the first part of our topic on proof techniques. A proof is a sequence of connected logical statements that explains why a statement is true. This module is about learning to read and write such proofs, and about the three standard techniques you will use throughout the course:

- **Direct proof** (deductive proof): start from the premise, and use definitions, theorems, axioms and logic to arrive at the conclusion.
- **Proof by contradiction** (indirect proof): assume the statement is false, follow the logical consequences, and arrive at a contradiction; the statement must therefore be true.
- **Proof by contrapositive**: use the fact that the implication $A \to B$ is equivalent to $\neg B \to \neg A$, and prove the easier direction.

You may wonder why proofs are needed at all, when a computer can check a property for every number up to a very large limit. The catch is that a computer can never confirm that a property holds for *all* numbers, and it cannot tell us *why* things work. If you claim that the software you created works for every input, a proof is how you convince not only yourself but also your clients.

While you work through this worksheet, keep a small puzzle in the back of your mind: you have 27 marbles, all but one weighing the same, and you may use a traditional two-pan scale only three times. Can you find the light marble? Notice the single lighter marble hidden among the 27 and the three-weighing limit. We will revisit this puzzle at the end of the topic.

Estimated time: approximately 60 minutes.

How to use this worksheet. Work through the activities in order, from the most basic to the most involved. Make a genuine attempt at each activity on paper before you look at the answer sheet; the answers are fully worked out, but the learning happens in the attempt.

Key terms and notation

Term	Meaning
Proof	A sequence of connected logical statements that explains why a statement is true.
Theorem / claim	A statement that we want to prove true.
Direct proof	A proof that starts from the premise, uses definitions, theorems, axioms and logic, and arrives at the conclusion.

Term	Meaning
Proof by contradiction	An indirect proof: assume the statement is false, use logical steps, and arrive at a statement that contradicts the assumption; the statement is therefore true.
Contradiction	A statement that is logically impossible or conflicts with a previously established fact or assumption, such as obtaining both P and $\neg$P
Contrapositive	$A \to B$ is equivalent to $\neg B \to \neg A$; proving the contrapositive proves the original statement.
Implication $A \to B$	True in every case except when A is true and B is false.
Even	A number of the form $2k$ for some integer k.
Odd	A number of the form $2k + 1$ for some integer k.
Rational	A number that can be written as p/q for integers p and q with $q \neq 0$.
Simplest form	A fraction p/q that cannot be simplified any further.
Prime number	A number greater than 1 that is not divisible by any number except 1 and itself.
Multiple of 3	A number of the form $3k$ for some integer k.
$\mathbb{N}$	The set of natural numbers.
Premise	The if-part of an implication.
Inductive step	The step showing that if a statement holds for some k, it holds for $k + 1$.

Common pitfall. An even number is twice an *integer* ($n = 2k$), not twice an odd integer. The integer k may itself be even or odd.

Common pitfall. The negation of "if A then B" is not "if A then not B". An implication is false only when the premise is true and the conclusion is false, so its negation is "A is true and B is false".

Gotcha. When you multiply an inequality by a negative number, the inequality sign flips. This is the key step in proofs that involve negative numbers.

As a checklist for the three techniques: a direct proof starts by assuming the

premise is true; a proof by contradiction starts by assuming the statement is false; a proof by contrapositive starts by assuming the negation of the conclusion.

Activities

Activity 1

Start point. Direct proof. A direct proof starts from the premise of the statement and uses definitions, algebra and known facts (such as the closure of integers under addition) to reach the conclusion. There is no special technique to remember; the skill is in choosing the right first step, which is usually to unpack the definitions.

Instructions. Complete the direct proof of the theorem: *If n and m are both even, then $n + m$ is even.* Fill in the blanks (1) and (2), and write a short justification for each line of the proof.

Proof. Let n and m be even. By the definition of even, $n = 2k$ and $m = 2l$ for integers k and l. So $n + m = 2k + 2l =$ (1) _____. Integers are closed under addition, so $k + l$ is an integer; call it t. Therefore $n + m = 2t$, so $n + m$ is (2) _____.

Expected output. A completed proof: the two blanks filled in, plus one short justification per line (for example: definition of even; factorising out 2; closure of integers under addition; definition of even).

Answers: see Activity 1 in the answer sheet. Do not look before making a proper attempt.

Activity 2

Start point. Direct proof with cases. When a statement must hold for all natural numbers, you may split the proof into cases and prove the statement in each case. A common split is even versus odd, because every natural number is one or the other.

Instructions. Prove: *For all $n \in \mathbb{N}$, $n^2 + n$ is even.* Case 1 is started for you; complete Case 2.

Case 1: n is even. Then $n = 2k$ for some integer k, so $n^2 + n = (2k)^2 + 2k = 4k^2 + 2k = 2(2k^2 + k)$, which is even.

Case 2: n is odd. Then $n = 2k + 1$ for some integer k. Expand $n^2 + n$, collect like terms, and factor out 2 to show the result is even.

> **Hint.** Expand $(2k + 1)^2$ carefully: it is $4k^2 + 4k + 1$, not $4k^2 + 1$.

Expected output. The completed odd case: the expansion $4k^2+4k+1+2k+1 = 4k^2 + 6k + 2$, the factorisation $2(2k^2 + 3k + 1)$, and the conclusion that $n^2 + n$ is even.

Optional extension (outside the time budget). Using the same case technique, prove that for all $x \in \mathbb{N}$, $2x^3 + x$ is a multiple of 3, by considering the three cases $x = 3k$, $x = 3k + 1$ and $x = 3k + 2$.

Answers: see Activity 2 in the answer sheet. Do not look before making a proper attempt.

Activity 3

Start point. Proof by contradiction. To prove a statement A by contradiction, assume A is false, then use definitions and logical steps until you reach a statement that contradicts your assumption. The assumption must be wrong, so A is true.

Instructions. Complete the proof that $\sqrt{2}$ is irrational by filling in the blanks (1), (2) and (3).

Proof. Assume, for contradiction, that $\sqrt{2}$ is rational. Then $\sqrt{2} = p/q$ for integers p and q with $q \neq 0$, and we take p and q in their simplest form. Squaring both sides gives $2 = p^2/q^2$, so (1) _____. Equation (1) says that p^2 is twice an integer, so p^2 is even and therefore p is even: $p = 2k$ for some integer k. Substituting into (1): $(2k)^2 = 2q^2$, so $4k^2 = 2q^2$ and hence $q^2 = 2k^2$. So q^2 is twice an integer, which means (2) _____. But if p and q are both even, then 2 is a common divisor and p/q could have been simplified, contradicting our choice of simplest form. Therefore the assumption was false, so (3) _____.

Expected output. (1) $p^2 = 2q^2$; (2) q is even; (3) $\sqrt{2}$ is irrational.

Answers: see Activity 3 in the answer sheet. Do not look before making a proper attempt.

Activity 4

Start point. Proof by contradiction with an existence claim. The same technique works for statements about what exists: assume the opposite (here, that only finitely many primes exist) and derive a contradiction.

Instructions. Write, in your own words, a proof by contradiction that there is an infinite number of prime numbers, following these prompts. Use the prompts as your proof skeleton; a short labelled answer is enough.

(a) What do you assume at the start?
(b) How do you form the number N from the assumed list?
(c) What is the remainder when N is divided by any prime p_i in the list?
(d) What does that tell you about N?
(e) Why does this contradict your assumption?

Expected output. Five labelled answers forming a short proof: assume the primes are exactly $p_1, p_2, \ldots, p_n$; set $N = p_1 p_2 \cdots p_n + 1$; the remainder is 1; so N is not divisible by any prime in the list and has a prime factor not in the list; this contradicts the assumption that the list contained all primes.

Answers: see Activity 4 in the answer sheet. Do not look before making a proper attempt.

Activity 5

Start point. Proof by contrapositive. The implication $A \to B$ is equivalent to its contrapositive $\neg B \to \neg A$, so proving the contrapositive proves the original statement. When the direct route is awkward, the contrapositive is often easier.

Instructions.

(a) State the contrapositive of the claim: *If $5n + 2$ is even, then n is even.*
(b) Prove the contrapositive by a direct proof: assume n is odd, write $n = 2k+1$, and simplify $5n + 2$ to show it is odd.
(c) State the contrapositive of: *For all $n \in \mathbb{N}$, if $n^3 + 1$ is odd, then n is even.*
(d) In one sentence, describe what an inductive step must show.

> **Hint.** In the contrapositive, both the premise and the conclusion are negated and swapped. For an integer, "not even" means "odd".

Expected output. (a) "If n is odd, then $5n + 2$ is odd"; (b) $5n + 2 = 5(2k + 1) + 2 = 10k + 7$, which is odd; (c) "If n is odd, then $n^3 + 1$ is even"; (d) It shows that truth for k implies truth for $k + 1$.

Answers: see Activity 5 in the answer sheet. Do not look before making a proper attempt.

Wrap-up

In this worksheet you have written a direct proof from definitions, carried out a case analysis, completed two proofs by contradiction (the irrationality of $\sqrt{2}$ and the infinitude of primes), and proved a statement by its contrapositive. You should now be able to follow a sequence of justified steps to a conclusion, prove a statement by first assuming it is false, and recognise when the contrapositive offers an easier route. These are the building blocks of formal proof. In the next part of the topic we will add proof by induction, and we will return to the 27-marble puzzle, where the case-splitting ideas you have practised here will help you plan the three weighings.

Answer sheet

Activity 1 (answer)

Completed proof, with justifications:

- Let n and m be even. (Assumption from the premise.)
- By the definition of even, $n = 2k$ and $m = 2l$ for integers k and l. (Definition of even.)
- $n + m = 2k + 2l = 2(k + l)$. (Factorising out 2.)

- Integers are closed under addition, so $k + l$ is an integer; call it t. (Closure of integers under addition.)
- $n + m = 2t$, so $n + m$ is even. (Definition of even.)

So (1) is $2(k + l)$ and (2) is *even*.

Gotcha. The conclusion "even" is valid only once you have the form 2 times an integer. Writing $2k + 2l$ and stopping there does not show the sum is even; you must factor out the 2 and note that $k + l$ is an integer.

Activity 2 (answer)

Case 1 (given): $n = 2k$, so $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, which is even.

Case 2: $n = 2k + 1$. Then

$$n^2+n = (2k+1)^2+(2k+1) = 4k^2+4k+1+2k+1 = 4k^2+6k+2 = 2(2k^2+3k+1),$$

which is even. Since every natural number is either even or odd, $n^2 + n$ is even for all $n \in \mathbb{N}$.

Gotcha. The most common error is expanding $(2k + 1)^2$ as $4k^2 + 1$ and losing the middle term $4k$. The two constant 1's (one from the square, one from the $+n$) combine to 2, which is exactly what supplies the factor 2.

Optional extension (answer). Factor first: $2x^3 + x = x(2x^2 + 1)$.

- $x = 3k$: $x(2x^2 + 1) = 3k(2x^2 + 1)$, a multiple of 3.
- $x = 3k+1$: $x(2x^2+1) = (3k+1)(18k^2+12k+3) = 3(3k+1)(6k^2+4k+1)$, a multiple of 3.
- $x = 3k+2$: $x(2x^2+1) = (3k+2)(18k^2+24k+9) = 3(3k+2)(6k^2+8k+3)$, a multiple of 3.

Every natural number falls into one of these three cases, so $2x^3 + x$ is a multiple of 3 for all $x \in \mathbb{N}$.

Activity 3 (answer)

(1) $p^2 = 2q^2$.
(2) q is even.
(3) $\sqrt{2}$ is irrational.

Full proof: assume $\sqrt{2}$ is rational, so $\sqrt{2} = p/q$ in simplest form with $q \neq 0$. Squaring both sides gives $2 = p^2/q^2$, so $p^2 = 2q^2$. Then p^2 is even, so p is even: $p = 2k$ for some integer k. Substituting, $(2k)^2 = 2q^2$, so $4k^2 = 2q^2$ and $q^2 = 2k^2$, which means q is even. But then p and q share the common divisor 2, so p/q was not in simplest form — a contradiction. The assumption was false, so $\sqrt{2}$ is irrational.

Gotcha. The contradiction is with the "simplest form" condition, not with the equation $p^2 = 2q^2$ itself. Forgetting to state the simplest-form assumption at the start is the most common way this proof breaks down.

Activity 4 (answer)

(a) Assume there are only finitely many prime numbers; call them $p_1, p_2, \ldots, p_n$. We are assuming this is the complete list of all primes.
(b) Form $N = p_1 p_2 \cdots p_n + 1$, the product of all the primes in the list plus one.
(c) Dividing N by any p_i leaves remainder 1, because N is one more than a multiple of p_i.
(d) So N is not divisible by any prime in the list. Since every integer greater than 1 has a prime factor, N must have a prime factor that is not in the list.
(e) This prime factor is a prime number that is not in the list $p_1, \ldots, p_n$, contradicting the assumption that the list contained all primes. Therefore there is an infinite number of prime numbers.

Gotcha. The "+1" is essential: it is what forces the remainder to be 1. If you used the product alone, every p_i would divide it and the argument would fail. The key point is that N cannot be built from the primes in the list, so the list was incomplete.

Activity 5 (answer)

(a) Contrapositive: *If n is odd, then $5n + 2$ is odd.*

(b) Assume n is odd, so $n = 2k + 1$ for some integer k. Then

$$5n + 2 = 5(2k + 1) + 2 = 10k + 5 + 2 = 10k + 7.$$

The term $10k$ is even, and adding 7 (odd) gives an odd number; indeed the units digit of $10k + 7$ is always 7. Hence $5n + 2$ is odd, and the contrapositive is proved. Since the contrapositive is equivalent to the original claim, "if $5n + 2$ is even, then n is even" is true.

(c) Contrapositive: *If n is odd, then $n^3 + 1$ is even.*

(d) It shows that truth for k implies truth for $k + 1$.

Gotcha. Do not confuse the contrapositive with a proof by contradiction. The contrapositive of "if A then B" is "if not B then not A" — both sides are negated and swapped. In part (b) you are doing a direct proof of the contrapositive: you never assume the original claim is false.